MCS-021

1. What is a stack ? Explain PUSH and POP operations of stack with the help of algorithms for each operation.
2. What is a Circular Queue ? Write an algorithm for insertion and deletion in a Circular Queue.
3. Write Kruskal's algorithm. Find the Minimum Cost Spanning Tree of the graph given below using Kruskal's algorithm:
4. What is a sparse matrix ? Write an algorithm that accepts a 6 × 5 sparse matrix and output 3-tuple representation, of the matrix.

5. Write an algorithm for the implementation of a Singly Linked List.
   Write an algorithm to implement Doubly Linked List.
6. Write an algorithm to implement the insertion and deletion operations into a singly linked list.
   Write an algorithm to insert and delete a node from a doubly-linked list.

7. What is a spanning tree ? What are its applications ? Write Kruskal's algorithm to find minimum cost spanning tree and explain it in terms of its complexity.
8. What is an AVL tree ? Explain the balancing methods of an AVL tree with an example. Explain the types of rotations performed on AVL trees with an example of each.
9. What is splay tree ? How is it different from a tree ? Explain.
10. What is a RED-BLACK tree ? Explain the properties of Red-Black Tree. Explain the procedure to insert elements in a RED-BLACK tree with the help of an example.

11. What is a linear search ? Write linear search algorithm and find its time complexity.
12. What is a binary search ? Write an algorithm for binary search and find its complexity.

13. Explain Quick Sort algorithm and trace the algorithm for the following set of data : 25, 0, 8, 4, 6, 18, 28
14. Write the Bubble Sort Algorithm. Find its time complexity and sort the following set of data using Bubble Sort : 8, 4, 2, 9, 18
15. Sort the following set of data using Insertion sort : 10 25, 15, 10, 18, 12, 4, 17

16. Write the steps involved to find the shortest path from Vertex '1' of the graph given below by using Dijkstra's algorithm :
17. Write short notes on the following :
    (i) Sequential File Organization
    (ii) Indexed Sequential File Organization